BECE MOCK EXAM
MATHEMATICS 2&1
Essay and Objective
2 hours

2&1

Name: .....

Index Number: .....

Basic Education Certificate Mock Examination

MATHEMATICS 2 & 1 — 2 hours

Essay and Objective

Do **not** *open this booklet until you are told to do so. While you are waiting, read and observe the following instructions. Write your* **name** *and* **index number** *in* **ink** *in the spaces provided above.*

This booklet consists of two papers. Answer Paper **2** *which comes first, in your answer booklet and Paper* **1** *on your Objective Test answer sheet. Paper* **2** *will last 1 hour after which the answer booklet will be collected. Do* **not** *start Paper* **1** *until you are told to do so. Paper* **1** *will last 1 hour.*

The use of calculators is **not** *allowed.*

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PAPER 2

ESSAY

Answer **four** *questions* **only**.

All questions carry equal marks.

All working must be clearly shown. Marks will **not** *be awarded for correct answers without corresponding working.*

1. (a) If T = {Prime numbers less than 11} and W = {Odd numbers less than 11},
   (i) List the members of $T \cap W$;
   (ii) List all the subsets of $T \cap W$.

   (b) Four siblings shared an amount of money as follows : Abena had $^1/_5$ of the amount and the remaining is shared among Ama, Kojo and Kofi in the ratio 1 : 2 : 3 respectively. If Kofi had GH¢ 2,400.00, how much did they share?

2. (a) Solve $7 - 3x \geq x + 3$ where x is a real member. Illustrate your result on the number line.

   (b) Mr Kumi spent GH¢ 2,100.00 of his salary on food, rent and transportation. He spent 15% of the salary on utility bills and saved the remaining. If the monthly salary for Mr Kumi is GH¢ 3,500.00,
   (i) what percentage of the salary is spent on food, rent and transportation;
   (ii) find the amount of money saved.

   (c) The equation of a straight line is given as $8x - 4y = 2$.
   (i) Find the slope of the straight line.
   (ii) Write the coordinate for the point where the line intercepts the y - axis.

3. (a) A trader sold 240 mangoes at 8 for GH¢ 3.20. If she bought the mangoes at GH¢ 80.00,
   (i) find the selling price of the 240 mangoes;
   (i) calculate the percentage profit the trader made.

   (b) Find the truth set of $^2/_5[(m - 1) - (2m - 2)] = {}^3/_5$.

   (c) Given that $r = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$ and $t = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, calculate $^1/_2(r + 2t)$.

4. (a) Using a ruler and a pair of compasses **only**, construct
   (i) triangle ABC such that $|AB| = 8$ cm, angle $ABC = 90^0$ and angle $BAC = 30^0$.
   (ii) the bisector of angle ACB.

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(b) The length of a rectangle is $(2n+12)$ metres and the breadth is $(n+3)$ metres. If the perimeter of the rectangle is 90 m.

(i) find the value of n;

(ii) calculate the area of the rectangle.

(c) Subtract $2y^2$ from $(2y)^2$ where $y = 3$.

5. (a) The sum of the ages of Esi and Araba is 28 . If Esi's age is two-fifths Araba's age, find their ages.

(b) Evaluate $\frac{3500 \times 0.035}{0.5}$, leaving your answer in standard.

(c) A shop gives a discount of 5% on all items bought . Mansa bought 5 wrist watches at GH¢ 80.00 each and 6 story books at GH¢ 25.00 each. How much should Mansa pay for the items when the discount is given?

6. (a) The following are marks obtained by 30 learners who wrote a test in Mathematics.

6	4	5	3	1	2
5	4	3	1	2	1
4	5	4	1	1	4
3	5	6	2	3	5
1	6	6	5	6	2

(i) Construct a frequency distribution table for the data;

(ii) Find the medium mark;

(iii) Calculate the mean mark.

(b) Two fair dice are tossed at the same time.

(i) Draw a sample space for the possible outcome.

(ii) Find the probability of the same number on the two dice.

END OF ESSAY TEST

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Answer all **the** *questions.*

Each question is followed by four options lettered A to D. Find the **correct** *option for each question and shade* **in pencil** *on your answer sheet the answer space which bears the same letter as the option you have chosen. Give only* **one** *answer to* **each** *question. An example is given below.*

If $3n + 2 = 8$, find the value of n.

A. 10
B. 6
C. 3
D. 2

The correct answer is 2, which is lettered D and therefore answer space D would be shaded.

[A] [B] [C] [D]

The carefully before you shade the answer space; erase completely any answers you wish t change

Do all rough work on this question paper.

Now answer the following questions.

1. What is the least number that should be added to 563 to make a sum divisible by 9?

A. 5
B. 6
C. 4
D. 8

2. Convert $^{3}/_{25}$ to a percentage.

A. 12%
B. 15%
C. 16%
D. 20%

3. If $a : 27 = 6 : 18$, find a.

A. 4
B. 9
C. 51
D. 81

4. Write four million, eight hundred and three thousand, one hundred and twenty-five in figures.

A. 4830125
B. 4803125
C. 4083125
D. 4831125

5. Kofi want to buy a rope that measures 1,500 cm. The tape measure used for measuring the rope for Kofi is in metres. How many metres should be measured for Kofi?

A. 1.5 metres
B. 1.50 metres
C. 15 metres
D. 150,000 metres

6. Correct 1564 to **three** significant figures

A. 1560
B. 1570
C. 15640
D. 156.4

7. A flash light cost GH¢ 30.00. A sales girl is given 10% commission on the number of flash lights she sells. If the sales girl sold 40 flash lights in a day, what is her commission for that day?

A. GH¢ 220.00
B. GH¢ 100.00
C. GH¢ 150.00
D. GH¢ 120.00

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8. The longer side of a right-angle triangle is 10 cm. If the lengths of the other two sides are 6 cm and 8 cm, find the area of the triangle.
   A. 24 cm²
   B. 30 cm²
   C. 40 cm²
   D. 48 cm²

9. How many lines of symmetry has a rectangular shape?
   A. 4
   B. 2
   C. 1
   D. 3

10. Express $2^4/_5$% as a decimal numeral.
   A. 0.28
   B. 2.8
   C. 0.028
   D. 0.0028

11. Evaluate 15 - 3(5 - 7) - (6 + 4) - 28.
   A. -29
   B. -17
   C. 29
   D. 17

12. Robert received an interest of GH¢120.00 on an amount he deposited at a rate of 15% per annum for $2^2/_5$ years. How much did Robert deposit?
   A. GH¢ 333.33
   B. GH¢ 430.33
   C. GH¢ 330.33
   D. GH¢ 350.00

13. The equation of a straight line is given as $2x + 2y = 6$. Find the gradient of the line.
   A. 2
   B. -2
   C. -1
   D. 1

14. The quadrant of a circle is divided into three equal sectors. What is the angle of each sector?
   A. 60°
   B. 90°
   C. 120°
   D. 30°

15. The rule of a mapping is $x \rightarrow 3 - 2x$. Find the image of -3 in the mapping.
   A. -9
   B. 9
   C. -6
   D. 6

16. A shop gives a discount of 15% on all items bought from the shop. How much does a customer pay for an item marked GH¢ 860.00?
   A. GH¢ 731.00
   B. GH¢ 129.00
   C. GH¢ 631.00
   D. GH¢ 703.00

17. A bag of rice cost $(2x - y)$ Ghana Cedis and a bag of sugar cost $(3y - x)$ Ghana Cedis. Write an expression for buying 4 bags of sugar and 3 bags of rice altogether.
   A. $2x + 9y$
   B. $10x - 9y$
   C. $3x - 9y$
   D. $2x - 9y$

18. The bearing of a town P from another town R is 035°. What is the bearing of town R from town P?
   A. 205°
   B. 215°
   C. 225°
   D. 315°

19. Transportation fares were increased by 5%. How much should a passenger pay if the old fare is GH¢ 50.00?
   A. GH¢ 48.50

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B. GH¢ 55.00
C. GH¢ 45.00
D. GH¢ 52.50

20. Round 8921465 to the nearest hundred.
A. 8921000
B. 8921400
C. 8921460
D. 8921500

21. Given that $r = \begin{pmatrix} -3 + y \\ x - y \end{pmatrix}$ and $p = \begin{pmatrix} 6 \\ -2 \end{pmatrix}$, find the values of x and y if $r = p$.
A. $x = 1, y = -3$
B. $x = -1, y = 9$
C. $x = -1, y = -9$
D. $x = -3, y = 3$

22. A car travelled at a speed of 100 km/h. Find the time spent to travel a distance of 950 km.
A. 9 hours 5 minutes
B. 9 hours 30 minutes
C. 6 hours 30 minutes
D. 6 hours 5 minutes

23. Solve for n given that $\frac{1}{3}n - (2n - 1) = 4 - \frac{1}{2}n$.
A. $1\frac{4}{7}$
B. $2\frac{2}{7}$
C. $2\frac{4}{7}$
D. $-2\frac{4}{7}$

24. The areas of a square and a rectangle are in the ratio 2 : 5 respectively. If the area of the square is 49 cm², calculate the area of the rectangle.
A. 88.5 cm²
B. 122.5 cm²
C. 98 cm²
D. 135 cm²

25. What is the image of the point (-2, 5) under a reflection in the line $y = -1$?
A. (-2, -4)
B. (-2, 4)
C. (-2, 6)
D. (-2, -7)

26. An electric blender that cost GH¢ 150.00 was sold for GH¢ 180.00. What is the percentage profit?
A. 20%
B. 15%
C. 30%
D. 10%

27. Factorize completely the expression: $2x^2 + x - 2x - 1$.
A. $(x - 1)(2x - 1)$
B. $(x + 1)(2x - 1)$
C. $(x - 1)(2x + 1)$
D. $(x + 1)(2x + 1)$

28. There are 36 identical balls in a sack. 10 of the balls are red, twice the number of red balls are green and the rest are yellow. Find the probability of picking a yellow ball from the sack.
A. $\frac{2}{5}$
B. $\frac{3}{7}$
C. $\frac{1}{5}$
D. $\frac{1}{6}$

29. The point A(-3, 7) is a reflection in the x-axis. Find its image.
A. (-3, -7)
B. (-3, 7)
C. (-7, 3)
D. (3, -7)

30. Solve the inequality: $1\frac{1}{2}x - (x - 1) \leq 5 - (x + 2)$.
A. $\frac{4}{3}$
B. $\frac{16}{3}$
C. 4
D. 5

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31. Make V the subject of the relation: $\frac{1}{t} = \frac{1}{V} - \frac{1}{r}$.

A. $V = \frac{rt}{r+t}$

B. $V = \frac{tr}{r-t}$

C. $V = \frac{r-t}{tr}$

D. $V = \frac{r+t}{rt}$

32. Find the next number in the sequence : 15, 9, 3, - 3, ...

A. - 6
B. - 12
C. - 18
D. - 9

33. Find the interest on GH¢ 150.00 at a rate of 20% per annum for 5 years.

A. GH¢ 100.00
B. GH¢ 150.00
C. GH¢ 300.00
D. GH¢ 500.00

34. A kg of meat cost GH¢ 23.00 and a box contains 50 kg of such meat. What is the cost of 3 boxes of meat?

A. GH¢ 1,150.00
B. GH¢ 2,300.00
C. GH¢ 3,300.00
D. GH¢ 3,450.00

35. Evaluate $^1/_2$ of $(^3/_4 - ^2/_5) + ^1/_5$.

A. $^3/_8$
B. $^5/_8$
C. $^7/_{40}$
D. $^3/_5$

36. What is the Highest Common Factor (HCF) of 18, 24 and 32?

A. 3
B. 2
C. 6
D. 4

37. Write 0.000463 in standard form.

A. 4.63×10^{-3}
B. 4.63×10^{4}
C. 4.63×10^{3}
D. 4.63×10^{-4}

38. The mean of 3, 4, 6, 8, x and 3 is 5. Find the value of x.

A. 6
B. 4
C. 5
D. 8

Use the magic square to answer questions **39** *and* **40**.

8	m	6
	5	7
4	n	

39. Find the value of m.

A. 3
B. 2
C. 1
D. 0

40. What is the value of m + n?

A. 12
B. 10
C. 15
D. 13

END OF ESSAY TEST

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